3.1 Law of Sines

Name _____ Period ____ Date _____

Pre-Calculus 3.1 · Student sheet

I can…

- use the Law of Sines to solve oblique triangles (AAS or ASA).
- use the Law of Sines to solve oblique triangles (SSA), including the ambiguous case.
- find the area of an oblique triangle and solve real-life problems.

Vocabulary · fill in the missing word

Oblique triangle — a triangle with no _____ angle

Law of Sines — a/sin A = _____ = c/sin C

Ambiguous case (SSA) — two sides and an angle _____ one of them: there may be no triangle, one, or two

Area of an oblique triangle — Area = _____ = ½ab sin C = ½ac sin B

Part 1 · Ratio check

Draw a large triangle with NO right angle. Label it ABC (side a across from angle A, and so on). Measure the sides in centimeters and the angles in degrees.

	Side (cm)	Opposite angle	side ÷ sin(angle)
a		A =	
b		B =	
c		C =	

What do you notice? Compare with another pair.

Part 2 · Notes and examples (fill in each note, then work the example)

Law of Sines: _____ = _____ = _____

When to use it: Use the Law of Sines for _____, _____, and _____.

Example 1

Solve the triangle: C = 102.4°, B = 28.7°, b = 27.4 feet. Round to two decimal places.

Example 2

Solve the triangle: $A = 22°$, $a = 12$, $b = 5$. Round to two decimal places.

The ambiguous case (SSA): In the SSA case, if $\sin B > 1$ there is _____. If $\sin B < 1$, check whether _____ also works.

Example 3

Show that there is no triangle with $A = 85°$, $a = 15$, and $b = 25$.

Example 4

Find two triangles with $A = 20.5°$, $a = 12$, and $b = 31$. Round to two decimal places.

Area of an oblique triangle = _____

Example 5

Find the area of a triangular lot with sides of 52 meters and 90 meters and an included angle of 102°.

Example 6

To find the distance from point A to a tree at point C across a river, a surveyor walks 233 meters along the bank from A to B. She measures $\angle CAB = 54°$ and $\angle CBA = 76°$. Find AC.

Part 3 · Partner practice

Partner 1

Solve the triangle: $A = 40°$, $B = 60°$, $a = 10$. Round to two decimal places.

Partner 2

How many triangles have $A = 30°$, $a = 6$, and $b = 10$? Find B for each.